11.2 Vectors

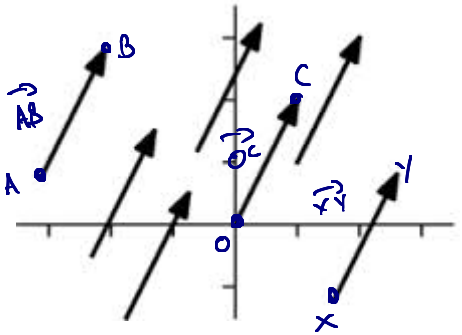
There are two types of quantities people measure: **Scalar** and **Vector**

Scalar
Determined by its magnitude only.
Example: length, temperature, voltage and mass

Vector
Determined by its magnitude and its direction, and usually represented by a directed line segment.
Example: velocity, acceleration and force

Definitions **(Vector, Initial and Terminal Point, Length)**

A **vector** in the plane is a **directed line segment**. The directed line segment $\vec{AB}$ has **initial point** A and **terminal point** B; its **length** is denoted by $|\vec{AB}|$. Two vectors are **equal** if they have the same length and direction.



$|\vec{AB}| = |\vec{OC}| = |\vec{XY}|$

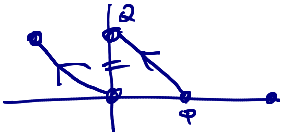
& they have the same directions.

So these are the same vectors.

Notation: $\mathbf{v}, \vec{v}, \overrightarrow{PQ} (\neq \overrightarrow{QP})$

Remark: A vector is measured by its length and direction, but not its position.

Given any directed line segment $\overrightarrow{PQ}$, we know there is a vector equal to $\overrightarrow{PQ}$ starting from the origin.



Definition:

If **v** is a **two-dimensional** vector in the plane equal to the vector with an initial point at the origin and terminal point (v_1, v_2), then the **component form** of **v** is $\mathbf{v} = \langle v_1, v_2 \rangle$

If **v** is a **three-dimensional** vector in the plane equal to the vector with initial point at the origin and terminal point (v_1, v_2, v_3), then the **component form** of **v** is

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle$$

For example, $P(1, 2, 3)$ is a point in space, while $\overrightarrow{OP} = \langle 1, 2, 3 \rangle$ is a vector in space.

If $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is represented by the directed line segment $\overrightarrow{PQ}$, where $P(x_1, x_2, x_3)$ is the initial point and $Q(y_1, y_2, y_3)$ is the terminal point, then $v_i = y_i - x_i$ for $i = 1, 2, 3$.

Eg. $P(1, 2, 3)$, $Q(4, 5, 6)$

$\overrightarrow{PQ} = \langle 4-1, 5-2, 6-3 \rangle = \langle 3, 3, 3 \rangle$

$= \overrightarrow{OQ} - \overrightarrow{OP}$

Definition: **Standard position vector** is the directed line segment with the initial point at the origin.

↳ initial point @ origin.

For example, consider the points $A(1, 1, 1)$ and $B(1, 2, 3)$ with the vector

$$\overrightarrow{AB} =< 0, 1, 2 >= \overrightarrow{OP}$$

Here P is the point $P(0, 1, 2)$ and $\overrightarrow{OP}$ is the standard position vector that is equal to $\overrightarrow{AB}$.

Definition: The **magnitude** or **length** of the vector $\mathbf{v} = \overrightarrow{PQ}$

If we have two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, then

$$|\overrightarrow{PQ}| = |\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}|$$

$$= |\langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle|$$

Example: Find the component form and length of the vector with initial point $P(-3, 4, 1)$ and the terminal point $Q(-5, 2, 2)$.

$$\overrightarrow{PQ} = \langle -5 - (-3), 2 - 4, 2 - 1 \rangle$$

$$= \langle -2, -2, 1 \rangle \leftarrow \text{component form}$$

$$|\overrightarrow{PQ}| = \sqrt{(-2)^2 + (-2)^2 + 1^2} = 3$$

Vector Algebra Operations

Two principal operations involving vectors are *vector addition* and *scalar multiplication*. A **scalar** is simply a real number.

Definition **Vector Addition and Multiplication of a Vector by a Scalar**

Let $\vec{u} =< u_1, u_2, u_3 >, \vec{v} =< v_1, v_2, v_3 >, k \in \mathbb{R}.$

Addition:

$$\vec{u} + \vec{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$$

Scalar Multiplications:

$$k\vec{u} = k\langle u_1, u_2, u_3 \rangle = \langle ku_1, ku_2, ku_3 \rangle$$

Vector Difference:

$$\vec{u} - \vec{v} = \vec{u} + (-\vec{v}) = \langle u_1 - v_1, u_2 - v_2, u_3 - v_3 \rangle .$$

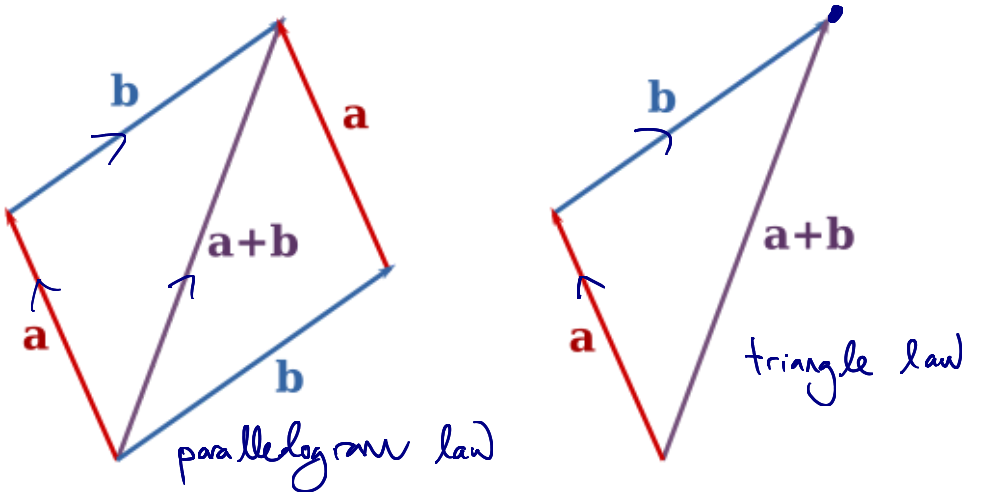
$-\vec{v}$: the vector in the opposite direction of $\vec{v}$

Vector Addition

Given two vectors **a** and **b**, the vector addition **a** + **b** can be illustrated geometrically in two ways.

1. **Parallelogram Law**
   If **a** and **b** are represented by two sides of a parallelogram, then **a** + **b** is a vector given by the diagonal of the parallelogram.

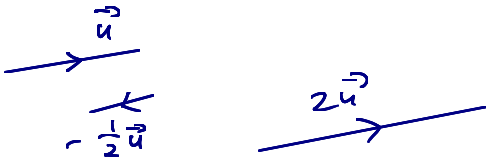
2. **Triangle Law**
   If **a** and **b** are represented by two sides of a triangle, then **a** + **b** is a vector given by the third side of the triangle.



Scalar Multiplication

Let c be a scalar and **v** a nonzero vector. Then

1. $|c\mathbf{v}| = |c||\mathbf{v}|$
2. Two vectors are parallel ⇔ one is scalar multiple of the others. That is, $\mathbf{u}\|\mathbf{v} \Leftrightarrow \mathbf{u} = c\,\mathbf{v}$ for some scalar c. In addition, u and v are in the same direction if $c > 0$; and in the opposite direction if $c < 0$.



Example:

1. $\mathbf{u} = < 3, 5, 4 >, \mathbf{v} = < -\frac{3}{2}, -\frac{5}{2}, -2 >$

$\vec{v} = -\frac{1}{2}\vec{u}$ so $\vec{u} \parallel \vec{v}$ but in opposite directions.

2. $\mathbf{u} = < 3, 5, 4 >, \mathbf{w} = < 3, 6, 0 >$

$\vec{u}$ is not parallel to $\vec{w}$ as we cannot write $\vec{u} = c\vec{w}$ for any $c \in \mathbb{R}$.

Example Performing Operations on Vectors

Let $\mathbf{u}=\langle -1, 3, 1 \rangle$ and $\mathbf{v}=\langle 4, 7, 0 \rangle$.

(a). $2\mathbf{u}+3\mathbf{v}$ $= \langle -2, 6, 2 \rangle + \langle 12, 21, 0 \rangle$

$= \langle 10, 27, 2 \rangle$

(b). $\mathbf{u}-\mathbf{v}$

$= \langle -5, -6, 1 \rangle$

(c). $\left|\frac{1}{2}\mathbf{u}\right|$ $= \left|\frac{1}{2}\right| |\vec{u}|$

$= \frac{1}{2}\sqrt{(-1)^2+3^2+1^2}$

$= \frac{\sqrt{11}}{2}$

Properties of Vector Operations

Vector operations have many of the properties of ordinary arithmetic.

Let $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ be vectors and a, b be scalars.

1. $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$ ← commutative
2. $(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$ ← associative.
3. $\mathbf{u}+\mathbf{0}=\mathbf{u}$ ← existence of identity
4. $\mathbf{u}+(-\mathbf{u})=\mathbf{0}$ ← inverse
5. $0\mathbf{u}=\mathbf{0}$
6. $1\mathbf{u}=\mathbf{u}$
7. $a(b\mathbf{u})=(ab)\mathbf{u}$ ← associative
8. $a(\mathbf{u}+\mathbf{v})=a\mathbf{u}+a\mathbf{v}$
9. $(a+b)\mathbf{u}=a\mathbf{u}+b\mathbf{u}$ ← distributive law

Unit Vectors

A vector **v** of length 1 is called a unit vector. The *standard unit vectors* are

$$\mathbf{i} = \langle 1, 0, 0 \rangle$$
$$\mathbf{j} = \langle 0, 1, 0 \rangle$$
$$\mathbf{k} = \langle 0, 0, 1 \rangle$$

Any vector $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ can be written as a linear combination of the standard unit vectors as follows:

$$\vec{v} = \langle v_1, v_2, v_3 \rangle = \langle v_1, 0, 0 \rangle + \langle 0, v_2, 0 \rangle + \langle 0, 0, v_3 \rangle$$
$$= v_1 \langle 1, 0, 0 \rangle + v_2 \langle 0, 1, 0 \rangle + v_3 \langle 0, 0, 1 \rangle$$
$$= v_1 \vec{i} + v_2 \vec{j} + v_3 \vec{k}$$

Remark: If $\vec{v} \neq \vec{0} \Rightarrow |\vec{v}| \neq 0$.

So $\left| \frac{1}{|\vec{v}|} \vec{v} \right| = \frac{1}{|\vec{v}|} |\vec{v}| = 1$

$\Rightarrow \frac{1}{|\vec{v}|} \vec{v}$ is a unit vector in the direction of $\vec{v}$.

Remark:
We can express any nonzero vector in terms of its two important features, its length and direction.

Given $\mathbf{v} \neq \mathbf{0}$, its length is $|\mathbf{v}|$ and unit vector in its direction is $\frac{1}{|\mathbf{v}|}\mathbf{v}$. So we can write

$$\mathbf{v} = |\mathbf{v}| \left(\frac{\mathbf{v}}{|\mathbf{v}|} \right).$$

length direction

Example: Find a unit vector **u** in the direction of the vector from $P_1(1, 0, 1)$ to $P_2(3, 2, 0)$.

$\vec{v} = \vec{P_1P_2} = \langle 2, 2, -1 \rangle$

$|\vec{v}| = \sqrt{2^2 + 2^2 + (-1)^2} = 3$

$\Rightarrow \quad \vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{\langle 2, 2, -1 \rangle}{3} = \langle \frac{2}{3}, \frac{2}{3}, -\frac{1}{3} \rangle$

Example:

1. Find the position vectors from $P(2, 3, 1)$ and $Q(-3, 0, 0)$. Find a unit vector in the direction of the vector from P to Q.

$$\vec{PQ} = \langle -3-2, 0-3, 0-1 \rangle = \langle -5, -3, -1 \rangle$$
$$= -5\vec{i} - 3\vec{j} - \vec{k}$$

$$|\vec{PQ}| = \sqrt{(-5)^2 + (-3)^2 + (-1)^2} = \sqrt{35}$$

A unit vector in the direction of $\vec{PQ}$ is

$$-\frac{5}{\sqrt{35}}\vec{i} - \frac{3}{\sqrt{35}}\vec{j} - \frac{1}{\sqrt{35}}\vec{k} = \left\langle -\frac{5}{\sqrt{35}}, -\frac{3}{\sqrt{35}}, -\frac{1}{\sqrt{35}} \right\rangle$$

2. Which of the following are unit vectors?

a. $\mathbf{u} = 3\mathbf{i} + \mathbf{j} - \mathbf{k}$

$|\vec{u}| = \sqrt{3^2 + 1^2 + (-1)^2} = \sqrt{11} \neq 1$

$\vec{u}$ not a unit vector

b. $\mathbf{v} = \frac{1}{2}\mathbf{i} - \frac{\sqrt{2}}{2}\mathbf{j} + \mathbf{k}$

$|\vec{v}| = \sqrt{\frac{1}{4} + \frac{2}{4} + 1} \neq 1$

$\vec{v}$ not a unit vector

c. $\mathbf{w} = \sin\theta\mathbf{i} - \cos\theta\mathbf{k}$

$|\vec{w}| = \sqrt{\sin^2\theta + \cos^2\theta}$

$= 1$

$\vec{w}$ is a unit vector for any θ.

3. Find a vector of magnitude 3 in the direction of the vector $\mathbf{v} = \frac{1}{2}\mathbf{i} + \frac{1}{2}\mathbf{j} - \mathbf{k}$.

The vector we are looking for is

$3\,\frac{\vec{v}}{|\vec{v}|}$

length = 1 ← direction

$|\vec{v}| = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \sqrt{\frac{3}{2}}$

$\Rightarrow \quad \therefore \quad 3\,\frac{\vec{v}}{|\vec{v}|} = \frac{3}{\sqrt{\frac{3}{2}}}\left(\frac{1}{2}\vec{i} + \frac{1}{2}\vec{j} - \vec{k}\right)$

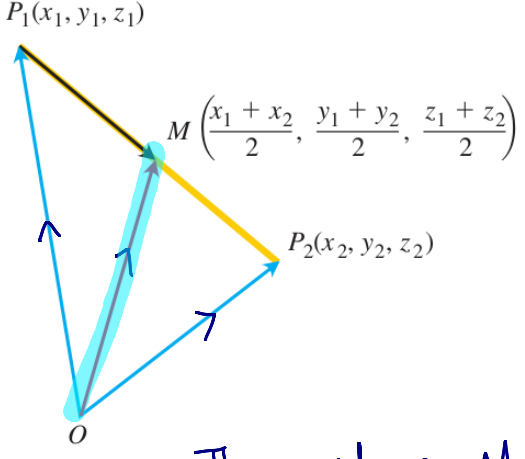
$= \sqrt{\frac{2}{3}}\left(\frac{1}{2}\vec{i} + \frac{1}{2}\vec{j} - \vec{k}\right)$

$= \left\langle \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\sqrt{\frac{2}{3}} \right\rangle$

The midpoint of a Line Segment

Suppose we have two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ in space. Let M be the midpoint of $\overrightarrow{P_1P_2}$.

$\vec{OP_1} = \langle x_1, y_1, z_1 \rangle$ $\quad \vec{P_1P_2} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$



$\vec{OM} = \vec{OP_1} + \vec{P_1M} \qquad \left(\vec{P_1M} = \frac{1}{2}\vec{P_1P_2}\right)$

$= \vec{OP_1} + \frac{1}{2}\vec{P_1P_2}$

$= \langle x_1, y_1, z_1 \rangle + \frac{1}{2}\langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$

$= \left\langle \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right\rangle$

The midpoint M has coordinate $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$.

E.g. The midpoint of the line segment joining $P_1(3, -2, 0)$ & $P_2(7, 4, 4)$ is $\left(\frac{3+7}{2}, \frac{-2+4}{2}, \frac{0+4}{2}\right) = (5, 1, 2)$.